Lab Number : 31 Date:2026/03/17

Title: Firewall Configuration

1.Install "firewall" package & start the service

Install command:

```
root@localhost:~# sudo yum install firewalld -y
Last metadata expiration check: 0:01:26 ago on Wed 18 Mar 2026 07:36:22 AM +0545
.
Package firewalld-2.4.0-1.el10.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Service start command:

```
root@localhost:~# sudo systemctl start firewalld
root@localhost:~# sudo systemctl enable firewalld
root@localhost:~# sudo systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset>
     Active: active (running) since Wed 2026-03-18 07:32:38 +0545; 6min ago
 Invocation: 3ba35f9c596b467b9737aa07ce1645fa
       Docs: man:firewalld(1)
   Main PID: 1210 (firewalld)
      Tasks: 2 (limit: 20120)
     Memory: 50.1M (peak: 50.4M)
        CPU: 912ms
     CGroup: /system.slice/firewalld.service
             └─1210 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Mar 18 07:32:36 localhost systemd[1]: Starting firewalld.service - firewalld - >
Mar 18 07:32:38 localhost systemd[1]: Started firewalld.service - firewalld - d>
lines 1-14/14 (END)
```

2. Add the following services 2 ports to allow packets through the firewall.
[service: http , smtp port = 25/tcp,25/udp,110/tcp]

Command:

```
root@localhost:~# sudo firewall-cmd --permanent --add-service=http
success
root@localhost:~# sudo firewall-cmd --permanent --add-service=smtp
success
root@localhost:~# sudo firewall-cmd --permanent --add-port=25/tcp
success
root@localhost:~# sudo firewall-cmd --permanent --add-port=25/udp
success
root@localhost:~# sudo firewall-cmd --permanent --add-port=110/tcp
success
root@localhost:~# sudo firewall-cmd --reload
success
```

Output:

```
root@localhost:~# sudo firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client http smtp ssh
  ports: 25/tcp 25/udp 110/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

3. Remove the following service & ports to block packets through the firewall. [service: 'smtp' port -25/tcp , 25/udp]

Command:

```
root@localhost:~# sudo firewall-cmd --permanent --remove-service=smtp
success
root@localhost:~# sudo firewall-cmd --permanent --remove-service=25/tcp
success
root@localhost:~# sudo firewall-cmd --permanent --remove-service=25/udp
success
root@localhost:~# sudo firewall-cmd --reload
success
```

Output:

```
root@localhost:~# sudo firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client http ssh
  ports: 25/tcp 25/udp 110/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```